10.3 Securing microservice communications

You've now deployed some APIs to Kubernetes and applied some basic security controls to the pods themselves by adding security annotations and using minimal Docker base images. These measures make it harder for an attacker to break out of a container if they find a vulnerability to exploit. But even if they can't break out from the container, they may still be able to cause a lot of damage by observing network traffic and sending their own messages on the network. For example, by observing communications between the Natter API and the H2 database they can capture the connection password and then use this to directly connect to the database, bypassing the API. In this section you'll see how to enable additional network protections to mitigate against these attacks.

10.3.1 Securing communications with TLS

In a traditional network, you can limit the ability of an attacker to sniff network communications by using *network segmentation*. Kubernetes clusters are highly dynamic, with pods and services coming and going as configuration changes, but low-level network segmentation is a more static approach that is hard to change. For this reason, there is usually no network segmentation of this kind within a Kubernetes cluster (although there might be between clusters running on the same infrastructure), allowing an